Business Requirements Document (BRD)

Project Title: AI-Powered FAQ Chatbot (RAG) for MPOnline Services

1. Executive Summary

The AI-Powered FAQ Chatbot is an intelligent system designed to provide accurate and instant responses to user queries related to MPOnline services. It uses Retrieval-Augmented Generation (RAG) to retrieve relevant data from documents and generate context-aware answers. The solution aims to reduce manual support workload, improve user experience, and provide a scalable support system with integrated grievance handling and analytics.

2. Business Objectives

- Provide instant and accurate answers to user queries
- Reduce dependency on manual customer support
- Improve accessibility of MPOnline service information
- Enable efficient grievance and feedback management
- Generate actionable insights through analytics

3. Stakeholders

- **End Users:** Citizens using MPOnline services
- **System Administrator:** Manages documents, users, and system operations
- **Support Experts:** Handle escalated queries
- **Organization:** MPOnline / Service Provider

4. Project Scope

In Scope:

- AI-based chatbot for answering FAQs
- Document upload and processing (PDF/Image)
- OCR-based text extraction
- Searchable knowledge base
- Expert escalation system ("Ask an Expert")
- Grievance registration and tracking system
- Feedback collection system
- Analytics dashboard

Out of Scope:

- Voice call-based support
- Offline functionality
- Integration with external government APIs (future enhancement)

5. Business Requirements

5.1 Chatbot Functionality

- The system shall allow users to ask questions through a chat interface
- The chatbot shall provide answers based on uploaded documents
- The chatbot shall maintain conversation history

5.2 Knowledge Base Management

- The system shall allow administrators to upload documents (PDF/Image)
- The system shall extract text using OCR
- The system shall index and store information for efficient retrieval

5.3 Expert Escalation

- The system shall provide an option to escalate queries to a human expert
- The system shall notify experts of pending queries
- The system shall allow tracking of responses

5.4 Grievance Management

- Users shall be able to submit complaints
- The system shall generate a unique ticket ID
- Users shall be able to track complaint status

5.5 Feedback System

- Users shall be able to rate responses
- The system shall store feedback for analysis

5.6 Analytics Dashboard

- The system shall display frequently asked questions
- The system shall identify unanswered queries
- The system shall track user activity and feedback trends

6. Non-Functional Requirements

- **Performance:** Response time should be less than 3 seconds
- **Scalability:** System should support increasing users and data
- **Security:** Secure handling of user data and queries
- **Usability:** User-friendly interface for both users and admins
- **Availability:** High system uptime and reliability

7. Business Process Flow

User Query → Retrieval → AI Response → Display → Escalation → Feedback

1. User submits a query via chatbot
2. System retrieves relevant information from knowledge base
3. AI generates a response
4. Response is displayed to the user
5. If unresolved → escalate to expert
6. User provides feedback

8. Technology Considerations

- **Backend:** Python (FastAPI)
- **Frontend:** React.js
- **AI Integration:** OpenAI API / Gemini API
- **OCR:** Tesseract
- **Database:** PostgreSQL / MongoDB

9. Success Criteria

- At least 80% of queries resolved by chatbot
- Average response time below 3 seconds
- User satisfaction rating above 4/5
- Reduction in manual support workload

10. Risks and Assumptions

Risks:

- OCR inaccuracies for scanned documents
- AI-generated incorrect responses (hallucination)
- Handling large document datasets

Assumptions:

- Users have internet access
- Documents provided are accurate and up-to-date
- Experts are available for escalation

11. Future Enhancements

- Multi-language support (Hindi + English)
- Voice-based interaction
- Mobile application integration
- Notification system for updates
- Self-learning AI improvements

12. Timeline

Phase	Duration
Requirement Analysis	1 Week
Development	2–3 Weeks
Testing	1 Week
Deployment	1 Week